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Master Functional Skills Model	Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P02

Past Paper 2

Time: 25 minutes | Paper Reference **PMAT2/N02**

Mathematics

Level 2
Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
16 /16

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

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- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and your answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used.**
- Take the value of π to be 3.14

Information

- The total mark for this section is 16.
- The marks for each question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 Here are the times, in seconds, five athletes took to finish a 100 m race.

11.05 10.94 11.12 10.91 11.12

(a) Find the median.

Step 1: Write all 5 values in ascending order (smallest to largest): *[M1 A1]* (2)

10.91, 10.94, 11.05, 11.12, 11.12

Middle value = Median (3rd out of 5)

Tutor note: With n = 5 items, middle position = (5 + 1) / 2 = 3rd item = 11.05

11.05 seconds

(b) Find the mode.

Definition: The mode is the most frequently occurring value in the dataset. *[B1]* (1)

- 10.91 appears 1 time
- 10.94 appears 1 time
- 11.05 appears 1 time
- 11.12 appears 2 times -> Highest frequency!

Therefore, the modal time is 11.12 seconds.

11.12 seconds

(Total for Question 1 is 3 marks)

2 Here is a formula

$$h = \sqrt{(a^2 - b^2)}$$

Find the value of h when

$a = 10$ and $b = 8$

(3)

Step 1: Substitute the given values a = 10 and b = 8 into the formula:

$$h = \sqrt{(a^2 - b^2)}$$

$$= \sqrt{(10^2 - 8^2)}$$

Step 2: Calculate each square (BIDMAS: Indices first):

- $10^2 = 10 \times 10 = 100$ *[M1]*
- $8^2 = 8 \times 8 = 64$

Step 3: Subtract under the square root:

$100 - 64 = 36$ *[M1]*

Step 4: Take the positive square root:

$h = \sqrt{36} = 6$ *[A1]*

Tutor Golden Rule: Sqrt($a^2 - b^2$) ≠ a − b ! Always calculate squares before subtracting.

Check: $6^2 + 8^2 = 36 + 64 = 100 = 10^2$ (Pythagorean triple: 6, 8, 10) (OK)

6

(Total for Question 2 is 3 marks)

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3 Alex is the manager of a hospital canteen.
He reviews the meals the patients choose.

On Monday there were 240 patients in total.

$\frac{1}{3}$ of these patients chose pasta.

$\frac{3}{8}$ of these patients chose beef stew.

The other patients chose chicken.

How many patients chose chicken on Monday?

(4)

Step 1: Find patients who chose pasta (1/3 of 240):

240 ÷ 3 = 80 patients *[M1]*

Step 2: Find patients who chose beef stew (3/8 of 240):

1/8 of 240 = 240 ÷ 8 = 30 patients

3/8 of 240 = 3 × 30 = 90 patients *[M1]*

Step 3: Total patients choosing pasta and beef stew combined:

80 + 90 = 170 patients

Step 4: Subtract from total patients (240) to find chicken:

Chicken = 240 − 170 = 70 patients *[M1 A1]*

Fraction check: 1/3 + 3/8 = 8/24 + 9/24 = 17/24 of patients.

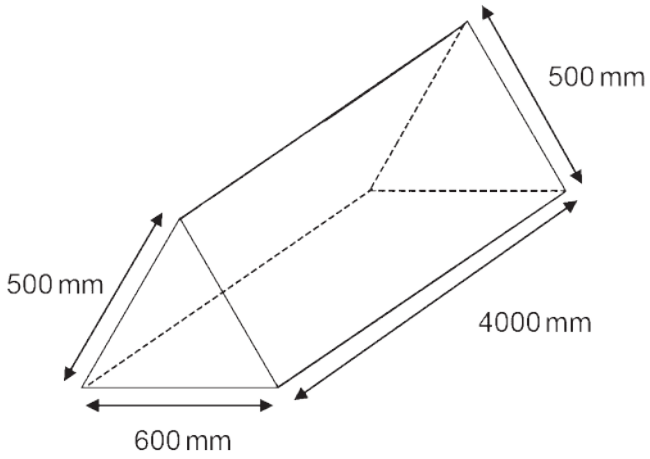
Remaining fraction choosing chicken = 1 − 17/24 = 7/24.

7/24 × 240 = 7 × 10 = 70 patients (OK)

70

(Total for Question 3 is 4 marks)

4 Roberto paints advertising boards.
Each board is in the shape of a triangular prism.



1 litre of paint covers 10 m^2
Roberto has 2 litres of paint.

Roberto has 3 advertising boards to cover with paint.
He needs to cover all five faces of the triangular prism with paint.

The area of each triangular face is 0.12 m^2

Will 2 litres of paint be enough to cover the 3 advertising boards?
You **must** show all your working.

(6)

Step 1: Convert all dimensions from mm to metres (divide by 1000):

- **Length = 4000 mm ÷ 1000 = 4.0 m** [M1]
- **Base of triangle = 600 mm ÷ 1000 = 0.6 m**
- **Sloping sides = 500 mm ÷ 1000 = 0.5 m each**

Step 2: Calculate the area of each face of the triangular prism:

- **2 Triangular ends = 2 × 0.12 m^2 = 0.24 m^2** [M1]
- **Bottom rectangle = 0.6 m × 4.0 m = 2.40 m^2** [M1]
- **2 Sloping sides = 2 × (0.5 m × 4.0 m) = 4.00 m^2**

Step 3: Total surface area of ONE board:

Total for 1 board = 2.40 + 4.00 + 0.24 = 6.64 m^2 [M1]

Working continued on next page ->

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Step 4: Total surface area to paint for ALL 3 boards:

Total area = 3 × 6.64 m² = 19.92 m² [M1]

```
  1 1   (carries)
  6.64
x    3
-------
19.92 m²
```

Step 5: Calculate paint coverage provided by 2 litres:

- 1 litre covers 10 m²
- 2 litres cover 2 × 10 m² = 20.00 m² [M1]

Step 6: Compare required area with available paint coverage:

19.92 m² < 20.00 m²

Surplus paint coverage = 20.00 − 19.92 = 0.08 m² remaining.

Conclusion: Yes, 2 litres is enough for all 3 boards. [A1]

Yes (19.92 m² < 20 m²)

(Total for Question 4 is 6 marks)

TOTAL FOR SECTION A IS 16 MARKS

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Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P02

Past Paper 2

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C02**

Mathematics

Level 2
Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
48 /48

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- Check your working and your answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used.**
- If your calculator does not have a π button take the value of π to be 3.14

Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
- The marks for each question are shown in brackets
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Advice

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- Check your answers if you have time at the end.

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Karen needs to buy a new fridge.
The fridge must fit in a space in the kitchen.

The space has width 19 inches, depth 22 inches and height 36 inches.

Karen sees these fridges for sale.

fridge	width in cm	depth in cm	height in cm
A	47.2	44.6	84.2
B	49.4	44.6	83.9
C	46.5	44.6	94.2

1 inch = 2.54 cm

Karen will buy one of these fridges.

Choose a suitable fridge for Karen to buy.
You **must** show your working.

(3)

Step 1: Convert maximum allowable space dimensions to cm (1 inch = 2.54 cm):

- Width = 19 in × 2.54 = 48.26 cm *[M1]*
- Depth = 22 in × 2.54 = 55.88 cm *[M1]*
- Height = 36 in × 2.54 = 91.44 cm

Step 2: Check each fridge against constraints (W ≤ 48.26, D ≤ 55.88, H ≤ 91.44):

Fridge A: Width 47.2 (OK), Depth 44.6 (OK), Height 84.2 (OK) -> FITS!

Fridge B: Width 49.4 cm > 48.26 cm -> Too wide, does NOT fit!

Fridge C: Height 94.2 cm > 91.44 cm -> Too tall, does NOT fit!

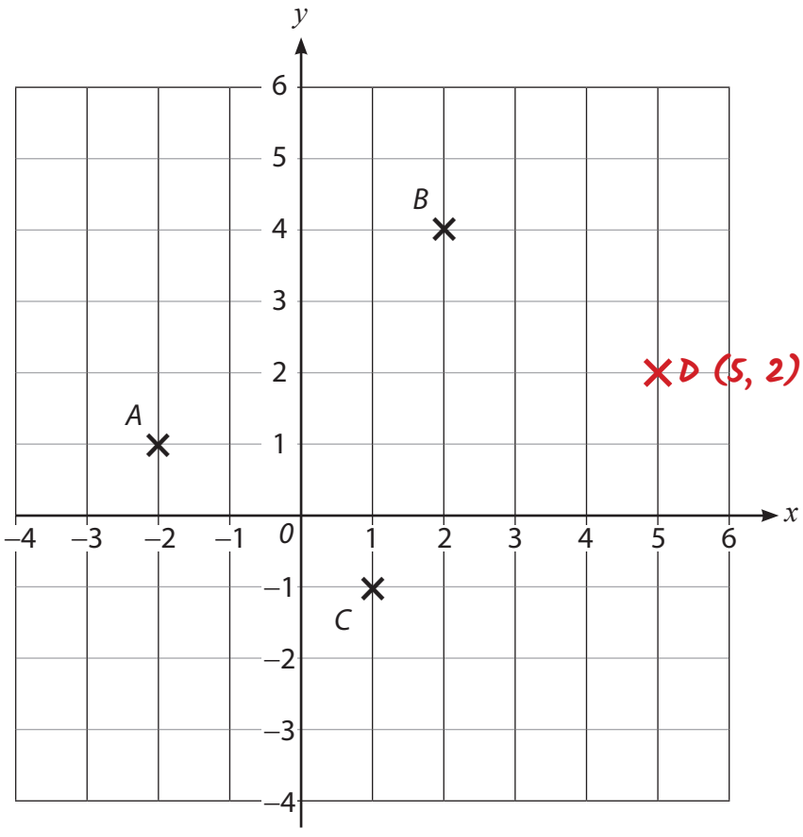
Conclusion: Only Fridge A can fit into the space. *[A1]*

Fridge A

(Total for Question 1 is 3 marks)

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2



(a) Write down the coordinates of point *C*.

Read coordinates: x = 1, y = −1 -> C(1, −1) *[B1]*

(1)

(1, −1)

(b) Mark with a cross the point *D* on the grid so that *ABCD* is a parallelogram.

Vector A(−2, 1) -> B(2, 4) is (+4, +3). Apply to C(1, −1): D = (1+4, −1+3) = (5, 2) *[B1]*

(1)

(c) Write down the sum of the angles in a parallelogram.

Rule: Sum of interior angles in any quadrilateral (parallelogram) = 360° *[B1]*

(1)

360 °

(Total for Question 2 is 3 marks)

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3 Carla is the director of a building company.
She employs builders at a site in Hull and at a site in London.

The average day rate of her builders in London is £153

In Hull the day rates she pays her builders are shown in this table

builder	A	B	C	D	E	F	G
day rate (£)	290	75	115	84	120	89	298

The builders in Hull say their average day rate is less than £153

Carla says the average day rate is the same in Hull and in London.

Show how both these statements can be true.
You **must** show your working.

(4)

Part 1: Calculate the MEAN day rate for builders in Hull (Carla's claim):

Sum = 290 + 75 + 115 + 84 + 120 + 89 + 298 = £1,071 *[M1]*

Mean = £1,071 ÷ 7 = £153 *[A1]*

-> Mean in Hull is £153 (same as London). Carla is correct!

Part 2: Calculate the MEDIAN day rate for builders in Hull (Builders' claim):

Order in Hull: 75, 84, 89, 115, 120, 290, 298 *[M1]*

4th of 7 in order is £115 -> Median = £115 *[A1]*

-> Median in Hull is £115 (< £153 London). Builders are correct!

Conclusion: Both statements are true using different averages:

- **Carla is correct using the MEAN average (£153).**
- **The builders are correct using the MEDIAN average (£115).**

Tutor note: Outliers (£290, £298) raise the mean, but median reflects typical pay.

(Total for Question 3 is 4 marks)

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4 Last year 123 students left a college with a pass in English.

This year 27 more students than last year left the college with a pass in English.

The college says

'The number of students leaving college with a pass in English has increased by 24% from last year.'

(a) Is the college correct?
Show why you think this.

(2)

Step 1: Calculate actual percentage increase in passes:

Formula: Percentage Increase = (Increase ÷ Original) × 100

% Increase = (27 ÷ 123) × 100 = 21.95% *[M1]*

Comparison & Decision (Alternative: 24% of 123 = 29.52 ≠ 27):

21.95% ≠ 24% (21.95% is less than 24%)

Conclusion: The college is NOT correct (No). *[A1]*

No (21.95% ≠ 24%)

☑ (b) Use a reverse calculation to show a check of your answer.

(1)

Step 2: Reverse calculation check:

Reverse check: 0.219512 × 123 = 27 ✓ *[B1]*

Reverse check confirms 27 students corresponds to ~21.95%, not 24% (OK)

(Total for Question 4 is 3 marks)

5 Tim did a survey at a large shopping centre.
He asked 400 visitors to the centre to choose the main reason for their visit.

The reasons were shops, free parking, food court and location.

(a) Complete the two-way table.

(3)

Step-by-step arithmetic to complete the two-way table:

1) Total Male = 400 − 227 = 173
2) Male Free Parking = 62 − 38 = 24 *[M1]*
3) Total Location = 21 + 23 = 44
4) Total Food Court = 400 − 198 − 62 − 44 = 96
5) Female Food Court = 96 − 40 = 56 *[M1]*
6) Male Shops = 173 − 24 − 40 − 21 = 88
7) Female Shops = 198 − 88 = 110 *[A1]*

Check: 110 + 38 + 56 + 23 = 227 (OK)

	shops	free parking	food court	location	total
male	88	24	40	21	173
female	110	38	56	23	227
total	198	62	96	44	400

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From the people taking part in the survey, one person is chosen at random to get a prize.

(b) Find the probability a female who chose location gets the prize.

(2)

Calculate probability a female who chose location gets the prize:

Formula: Probability = (Number of Successful Outcomes) ÷ (Total Outcomes)

- **Females who chose location = 23 (from table)** *[M1]*
- **Total survey participants = 400**

Probability = 23 / 400 = 0.0575 (or 5.75%) *[A1]*

Tutor note: 23 is prime, so 23/400 cannot be simplified further.

$$\frac{23}{400}$$

(Total for Question 5 is 5 marks)

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6 Wahab sells cars.

The table shows information about the number of cars he sold each week for the last 26 weeks.

number of cars sold each week	frequency	Midpoint (x)	f × x
1–5	2	3	2 × 3 = 6
6–10	12	8	12 × 8 = 96
11–15	9	13	9 × 13 = 117
16–20	3	18	3 × 18 = 54
	Total f = 26		Total fx = 273 [M1]

Wahab estimates he sold a mean average of 10 cars per week.

Is this estimate of the mean correct?
Show why you think this.

(3)

Step 1: Calculate the estimated mean cars sold per week:

Mean = Total fx ÷ Total f = 273 ÷ 26 = 10.5 [M1]

Step 2: Compare with Wahab's estimate of 10 cars:

10.5 ≠ 10 (Actual estimated mean is 10.5 cars, not 10)

Conclusion: Wahab is NOT correct (No). [A1]

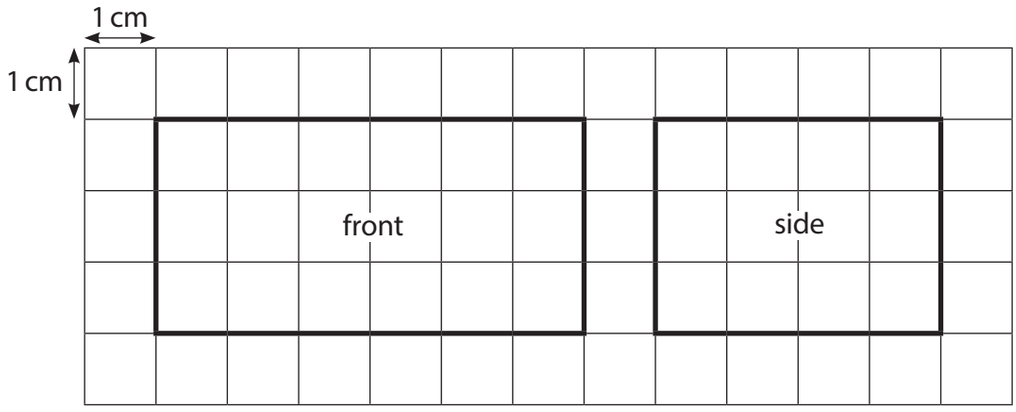
Alternative check: 26 weeks × 10 cars = 260 cars < 273 total cars (OK)

No (mean = 10.5)

(Total for Question 6 is 3 marks)

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7 The front elevation and the side elevation of a cuboid are drawn on the grid below.



(a) Find the volume of the cuboid.

(2)

Identify dimensions from elevations: Length = 6 cm, Width = 4 cm, Height = 3 cm

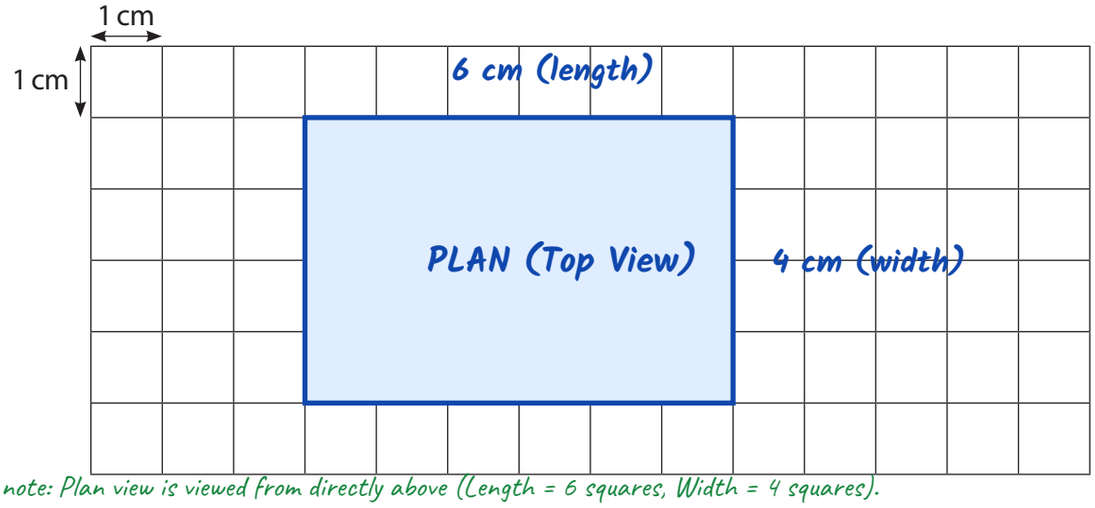
Volume = 6 cm × 4 cm × 3 cm = 72 cm³ *[M1 A1]*

72 cm³

(b) Draw the plan of the cuboid on the grid below.

[M1 A1]

(2)



Tutor note: Plan view is viewed from directly above (Length = 6 squares, Width = 4 squares).

(Total for Question 7 is 4 marks)

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8 Mina runs a football club.
She needs to order size 3, size 4 and size 5 footballs in the ratio 6 : 2 : 1

Mina needs to order a total of 180 footballs.

(a) Complete the order form for Mina.

(3)

Step 1: Find the total number of parts in the ratio:

Ratio = 6 : 2 : 1 -> Total parts = 6 + 2 + 1 = 9 parts

Step 2: Calculate the value of 1 part:

1 part = 180 footballs ÷ 9 = 20 footballs [M1]

Step 3: Calculate the quantity of each football size:

- Size 3: 6 parts = 6 × 20 = 120 footballs [M1]
- Size 4: 2 parts = 2 × 20 = 40 footballs
- Size 5: 1 part = 1 × 20 = 20 footballs [A1]

Check: 120 + 40 + 20 = 180 total footballs (OK)

order form	
size of football	**number**
3	120
4	40
5	20

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Mina also needs to buy medical equipment.

She needs 23 packets of plasters.

One packet costs £4.49

A box of 5 packets costs £20.25

(b) Can Mina buy 23 packets of plasters for less than £100?
Show why you think this.

(3)

Step 1: Find the cheapest combination to purchase 23 packets:

- 1 packet costs £4.49
- 1 box of 5 packets costs £20.25 (equiv. £4.05 per packet -> better value)

Divide: 23 ÷ 5 = 4 boxes with 3 individual packets remaining [M1]

Step 2: Calculate total cost for 4 boxes and 3 packets:

- 4 boxes of 5 = 4 × £20.25 = £81.00
- 3 individual packets = 3 × £4.49 = £13.47 [M1]

Total Cost = £81.00 + £13.47 = £94.47

Step 3: Compare with £100 budget:

£94.47 < £100.00

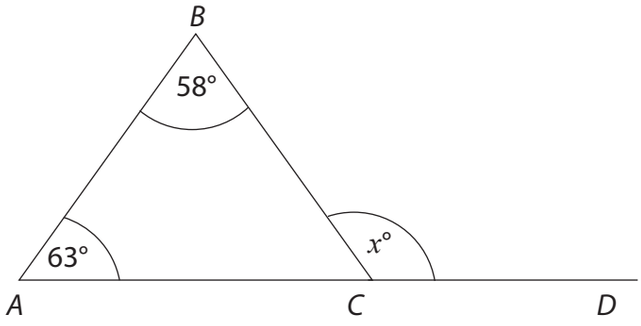
Remaining change = £100.00 – £94.47 = £5.53

Conclusion: Yes, Mina can buy 23 packets for < £100 [A1]

Yes (£94.47 < £100)

(Total for Question 8 is 6 marks)

9 *ABC* is a triangle.
ACD is a straight line.



(a) Work out the value of x.

(2)

Method 1 (Interior angles in triangle sum to 180°):

- **Triangle ABC: Angle BCA = 180° − 121° = 59°** *[M1]*
- **ACD is a straight line, so angles on a straight line sum to 180°:**

x = 180° − 59° = 121° *[A1]*

Method 2 (Exterior Angle Rule): Exterior angle = sum of two opposite interior angles:
x = 63° + 58° = 121°

121 °

☑ (b) Use estimation to check your answer.

(1)

Step 2: Estimation check (Round each angle to nearest 10°):

- **63° ≈ 60° and 58° ≈ 60°**
- **Estimated x ≈ 60° + 60° = 120°** ✓ *[B1]*

Estimate of 120° confirms the accuracy of 121° (OK)

(Total for Question 9 is 3 marks)

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10 There are roadworks on a motorway.
The average speed limit is 50 mph.

A car travels a distance of 18 miles in 24 minutes.

Is the average speed of the car less than 50 mph?
Show why you think this.

(3)

Step 1: Convert travel time from minutes to hours:

Rule: 60 minutes = 1 hour -> Divide minutes by 60

Time = 24 minutes ÷ 60 = 24/60 = 0.4 hours [M1]

Step 2: Calculate the average speed of the car:

Formula: Average Speed = Distance ÷ Time

Speed = 18 ÷ 0.4 = 45 mph [M1]

Step 3: Compare with the 50 mph speed limit:

45 mph < 50 mph

The car's average speed is 5 mph below the speed limit.

Conclusion: Yes, 45 mph is < 50 mph. [A1]

Alternative method: Distance covered at 50 mph in 24 minutes = 50 × 0.4 = 20 miles.
Since 18 miles < 20 miles, the car was slower than 50 mph (OK)

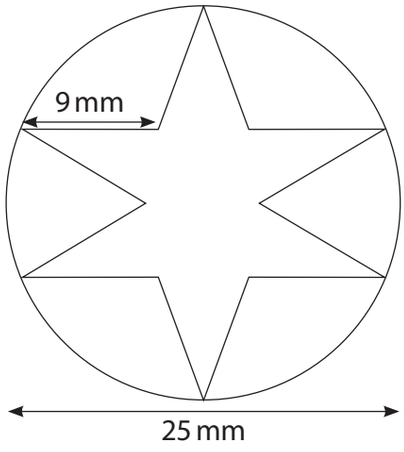
Yes (45 mph < 50 mph)

(Total for Question 10 is 3 marks)

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11 Mario uses wire to make earrings.

Mario uses wire to make a circle with diameter 25 mm.
He then adds a star shape made from wire.
All the sides of the star shape have length 9 mm.



Mario has 3 rolls of wire.
There are 500 mm of wire on each roll.

Mario is going to make 8 of these earrings.

Will Mario have enough wire to make 8 of these earrings?

(5)

Step 1: Calculate wire needed for CIRCLE (Circumference):

Formula: Circumference = pi × diameter

Circumference = π × 25 = 78.539... ≈ 78.54 mm *[M1]*

Step 2: Calculate wire needed for the STAR shape:

The 6-pointed star has 12 sides of 9 mm each:

Star perimeter = 12 × 9 mm = 108 mm *[M1]*

Step 3: Total wire needed for ONE earring:

Wire per earring = 78.54 + 108 = 186.54 mm *[M1]*

Working continued on next page ->

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Step 4: Total wire needed for ALL 8 earrings:

Total wire = 8 × 186.54 = 1492.32 mm [M1]

Step 5: Calculate total wire available on 3 rolls:

- 1 roll has 500 mm of wire
- Total available = 3 rolls × 500 mm = 1500 mm

Step 6: Compare wire needed with wire available:

1492.32 mm < 1500 mm

Surplus wire remaining = 1500 – 1492.32 = 7.68 mm left over.

Conclusion: Yes, Mario has enough wire. [A1]

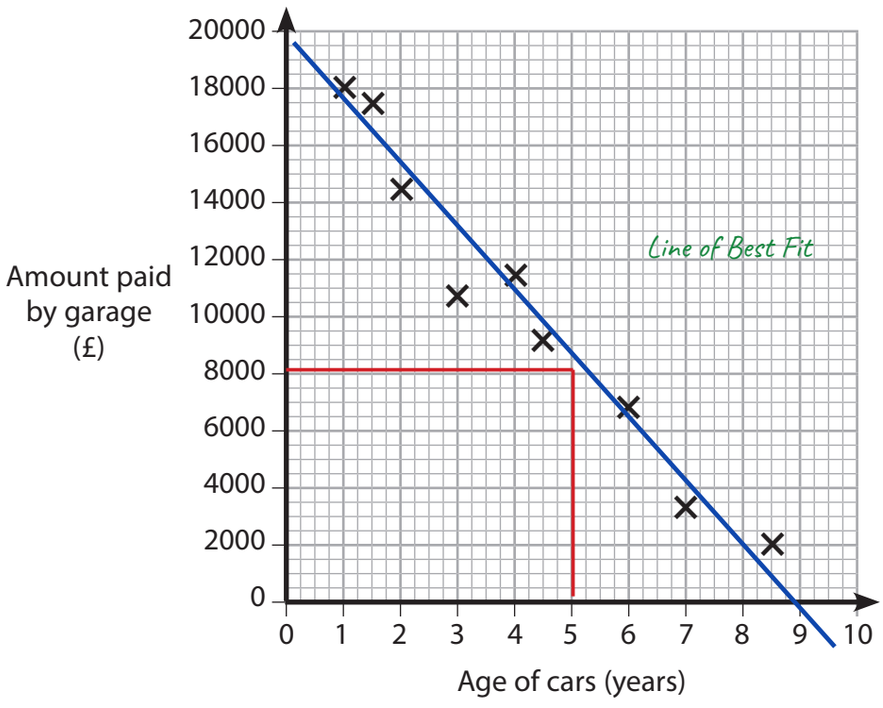
1492 mm < 1500 mm (7.68 mm surplus remaining)

Yes (1492 mm < 1500 mm)

(Total for Question 11 is 5 marks)

12 Daniel has a blog about how garages buy and sell cars.

The scatter diagram gives information about the amount garages pay for one type of car depending on its age.



Daniel is going to use the scatter diagram to estimate the monthly payments for a car the garage is going to sell.

Daniel knows garages sell cars for 34% more than they paid.

The car is 5 years old and there are 24 equal monthly payments.

Work out the cost of one monthly payment.

(6)

Step 1: Estimate amount garage paid for a 5-year-old car:

- *Draw a Line of Best Fit through the balance of points.*
- **At Age 5: Read across -> Estimated Price = £8,000** *[M1]*

(Acceptable mark scheme range for estimate: £7,500 to £8,500)

Working continued on next page ->

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Step 2: Calculate the selling price after a 34% increase on £8,000:

- **Markup: 34% of £8,000 = £2,720** [M1]
- **Selling Price = £8,000 + £2,720 = £10,720** [A1]

Multiplier method: Selling Price = £8,000 × 1.34 = £10,720

Step 3: Calculate the cost of one monthly payment over 24 equal installments:

Formula: Monthly Payment = Selling Price ÷ 24

Monthly Payment = £10,720 ÷ 24 = £446.66... [M1]

Step 4: Round to nearest penny (2 decimal places for money):

Round £446.66... -> £446.67 per month [A1]

Tutor note: The 3rd decimal digit is 6 (≥ 5), so round up to 67p.

Check: 24 × £446.67 = £10,720.08 ≈ £10,720 (OK)

£ **446.67**

(Total for Question 12 is 6 marks)

TOTAL FOR SECTION B IS 48 MARKS
TOTAL FOR PAPER IS 64 MARKS

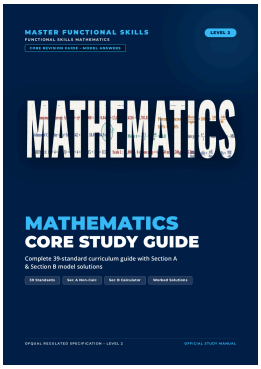
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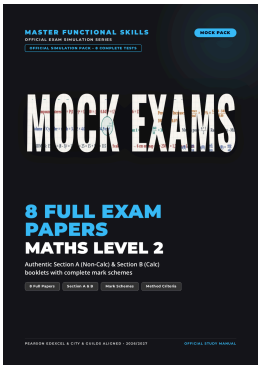


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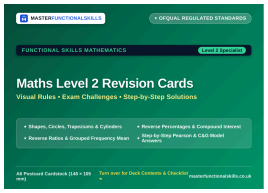


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